BV

HACKATHON ROUND 2 • UPDATED PROJECT REPORT

BRANDVAULT AI

Protect Every Word. Preserve Every Identity.

From proposal to working platform: what we built, how it works, and what comes next.

12	172	100	32
advanced intelligence engines shipped	automated tests, all passing	REST API endpoints	working application screens

TEAM

Deal Dynamos

TEAM ROSTER

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REPORT DATE

7 November 2026

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OPERATING PRINCIPLE

Agents advise — humans decide.
No autonomous publishing.

Supersedes: Round 1 Technical Proposal (3 November 2026)

All figures in this report are measured from the working codebase

CONTENTS

What this report covers

This report updates our Round 1 Technical Proposal. It covers the four items requested for Round 2, plus extra sections on quality, security, the demo and current limits.

01	Executive summary	Where the product stands today
02	List of completed features	Required item 1 · core workflow, 12 engines, platform
03	Product walkthrough	Screenshots of the running application
04	Technology & implementation details	Required item 2 · architecture, stack, AI pipeline, data
05	Testing & measured quality	Test suite and evaluation results
06	Security & responsible AI	How data and decisions are protected
07	Changes made to the original plan	Required item 3 · planned vs built, and why
08	Three major improvements for Round 3	Required item 4 · scope, approach, success measures, plan
09	Demo script & how to run	5-minute judge walkthrough
10	Known limitations	What is not done yet
11	Conclusion	

How to read status labels. Done working end to end and covered by tests · Partial works with stated limits · Not built not yet implemented · New added beyond the Round 1 plan.

01 · EXECUTIVE SUMMARY

A working pre-publication governance platform

In Round 1 we proposed BrandVault AI: a governance layer that checks every campaign against a company's approved brand knowledge before it is published. Every finding comes with evidence, and only an authorised person can approve. That platform now runs end to end. We also built 12 advanced intelligence engines that go well beyond the original MVP.

12 + 5
intelligence engines plus 5 core modules beyond the Round 1 plan

30
database tables, every row scoped to one organisation

0.925
F1 score on our 36-case evaluation set (precision 1.00)

13
agent types recording every run with timing

The governed path, as built

01	02 · HUMAN	03	04	05	06	07 · HUMAN	08
Upload brand guidelines	Admin confirms rules & facts	Submit campaign	Agents analyse with evidence	Product truth, claims, translation	Launch readiness check	Reviewer approves the exact version	Brand Passport (SHA-256)

What changed since Round 1

- The full MVP workflow works, from guideline upload to Brand Passport.
- Both Round 1 stretch goals are built: Risk Radar and Brand Memory.
- 12 new engines, from Product Truth and Launch Readiness to Fraud Intelligence.
- Translation Guard checks translated campaigns in 9 languages.
- Security hardening: tenant isolation, roles, version-locked approvals, rate limits, durable jobs.
- A professional light-theme interface with 32 screens.

Principles we kept

- **Deterministic first.** Prices, numbers, hashes and colour maths never depend on an AI guess.
- **Evidence or nothing.** A claim with no evidence is "Unsupported", never "false".
- **Humans decide.** No agent can approve, publish or change Brand DNA.
- **No invented data.** Every screen shows real records, and empty or thin data is labelled as such.

02 · REQUIRED ITEM 1

List of completed features

Every feature below runs against the real database and is covered by the automated test suite unless stated otherwise. The features are grouped into the core governance workflow promised in Round 1, the new modules, the 12 advanced engines and platform security.

A Core governance workflow (Round 1 MVP)

FEATURE	WHAT IT DOES	STATUS
Brand DNA Studio	Admin uploads PDF guidelines. Rules, approved facts and terminology are extracted and stay inactive until an administrator confirms them (human gate #1). Page-accurate citations come from the PDF page markers.	Done
Content Intelligence	Multi-agent analysis (Terminology, Voice & Tone, ClaimDNA, optional LLM Analyst, Suggestion). Gives a health score, findings with rule, evidence and source, and one-click corrections.	Done
ClaimDNA	The 5-class taxonomy from Round 1: Supported, Unsupported, Contradicted, Outdated, Needs Human Review. An Evidence Integrity Map separates "missing evidence" from "contradicted".	Done
Evidence retrieval (RAG)	Retrieval filtered by organisation and approval status. Hybrid ranking (vector cosine plus keyword) over persisted chunk embeddings, with page citations.	Done
Approval Hub	Human gate #2. Only Reviewers and Administrators can decide. Approvals are locked to the exact version and content hash; a stale decision is rejected (HTTP 409).	Done
Brand Passport	A SHA-256 certificate of the approved version with the real guideline version. It is superseded and re-issued when the content changes.	Done
Visual DNA Inspector	Measured colour palette, chromatic share and contrast for PNG/BMP creatives, saved as findings.	Partial
Agent Control Center	Every agent run is persisted with status and execution time (13 agent types currently recorded).	Done
Brand Risk Radar (R1 stretch)	Five explained risk indicators (content, claim, guideline, process, visual) with drivers and recommendations, computed from real history.	Done
Brand Memory (R1 stretch)	Learns from accepted corrections and recurring violations, and proposes new terminology. Only an administrator can apply a proposal.	Done

B The 12 advanced intelligence engines

Built in three phases. Each engine is integrated into an existing module rather than added as a separate product, and each states its own limits on screen.

#	ENGINE	WHAT IT DOES	WHERE
1	**Product Truth Engine**	Structured products and facts with source page, verbatim excerpt and validity dates. Verifies prices, battery, TOPS, GHz, RAM vs storage, display, weight and warranty. AI extraction proposes facts and discards any whose quote is not verbatim in the document.	Brand DNA
2	**Knowledge Gap Detector**	Completeness score plus a prioritised list of missing, conflicting, outdated and weakly-sourced brand knowledge, with recommended documents to upload.	Brand DNA
3	**Campaign DNA Fingerprint**	Verify any pasted copy against issued passports: Verified Match, Modified After Approval (with word diff), Unknown Version or New Unapproved Campaign. Uses an append-only log.	Brand Passport
4	**Asset Library + Forensics**	Official assets with SHA-256, pHash and dHash. Finds logos in an image (OpenCV ORB + RANSAC homography) and measures stretch, rotation, crop, recolour (ΔE) and outdated versions.	Visual DNA
5	**Campaign Launch Readiness**	12-point pre-launch checklist with 6 states, from Ready for Review to Expired Approval. Routes work to roles; never publishes or approves.	Approval Hub
6	**Asset Lifecycle & Expiry**	Lifecycle of assets, facts, offers, guidelines and approvals. Alerts on superseded assets still in use and campaigns quoting expired prices.	Visual DNA
7	**Brand Consistency Heatmap**	Findings by product, department and month, with compliance trends and filters. Demo data is labelled.	Command Center
8	**AI Reviewer Intelligence**	Compares AI findings with human decisions to find rules that may need tuning. Recommends only with repeated evidence. No reviewer rankings.	Approval Hub
9	**Campaign Stress Testing**	"What if?" scenarios on a copy of a campaign: price change, removed disclaimer, translation, discontinued product, older guideline. Real checks run in dry-run mode; nothing is saved to the campaign.	Brand Assistant
10	**Similarity Detector**	Near-duplicates, similar campaigns and reused sentences across campaigns, passports and translations, grouped into clusters. Image similarity via pHash.	Content Intelligence
11	**Brand Fraud Intelligence**	Investigates reported links, text or screenshots: look-alike domains (homoglyph, punycode, typosquat), unapproved prices, OTP and card requests, logo misuse. Numbered incidents with evidence and append-only history.	Risk Radar
12	**Reputation Early Warning**	Upload authorised feedback (CSV/JSON) to see sentiment, complaint categories, affected products, feedback disputing campaign claims, and trends.	Risk Radar

C New modules added beyond the plan

FEATURE	WHAT IT DOES	STATUS
Translation Guard	Compares translations with the source. Checks numbers, prices, "up to / more than" qualifiers, negations and brand terms. Full checks in 9 languages (en, hi, ta, fr, de, es, ja, zh, ar), plus an optional AI meaning score and corrected translation.	New
AI Copy Studio	Brand-compliant rewrite using the LLM, labelled with provider and model. Falls back to a deterministic rules-based rewrite when no model is configured.	New
Brand Assistant	Grounded question-and-answer over approved records with citations. Answers "not in approved data" for off-topic questions.	New
Knowledge Graph	Interactive graph of brand, categories, rules, findings, claims and passports built from real records.	New
Setup Guide & walkthrough	Onboarding driven by real completion status, plus a guided product tour with read-aloud narration.	New

D Platform, security & operations

Identity & roles

Email/password sign-in, plus Google sign-in when OAuth keys are configured. Three roles: Administrator, Reviewer and Creator. Bearer tokens signed with a managed secret key.

Tenant isolation

Every query is scoped to the signed-in organisation. Cross-organisation access is rejected, and this is covered by dedicated tests.

Durable jobs & limits

Background jobs stored in the database survive restarts. Rate limits protect sign-in and AI routes.

03 · PRODUCT WALKTHROUGH

The running application

All screenshots were captured from the live application running on the NOVA TECH demo workspace. They show real stored records, not mock-ups.

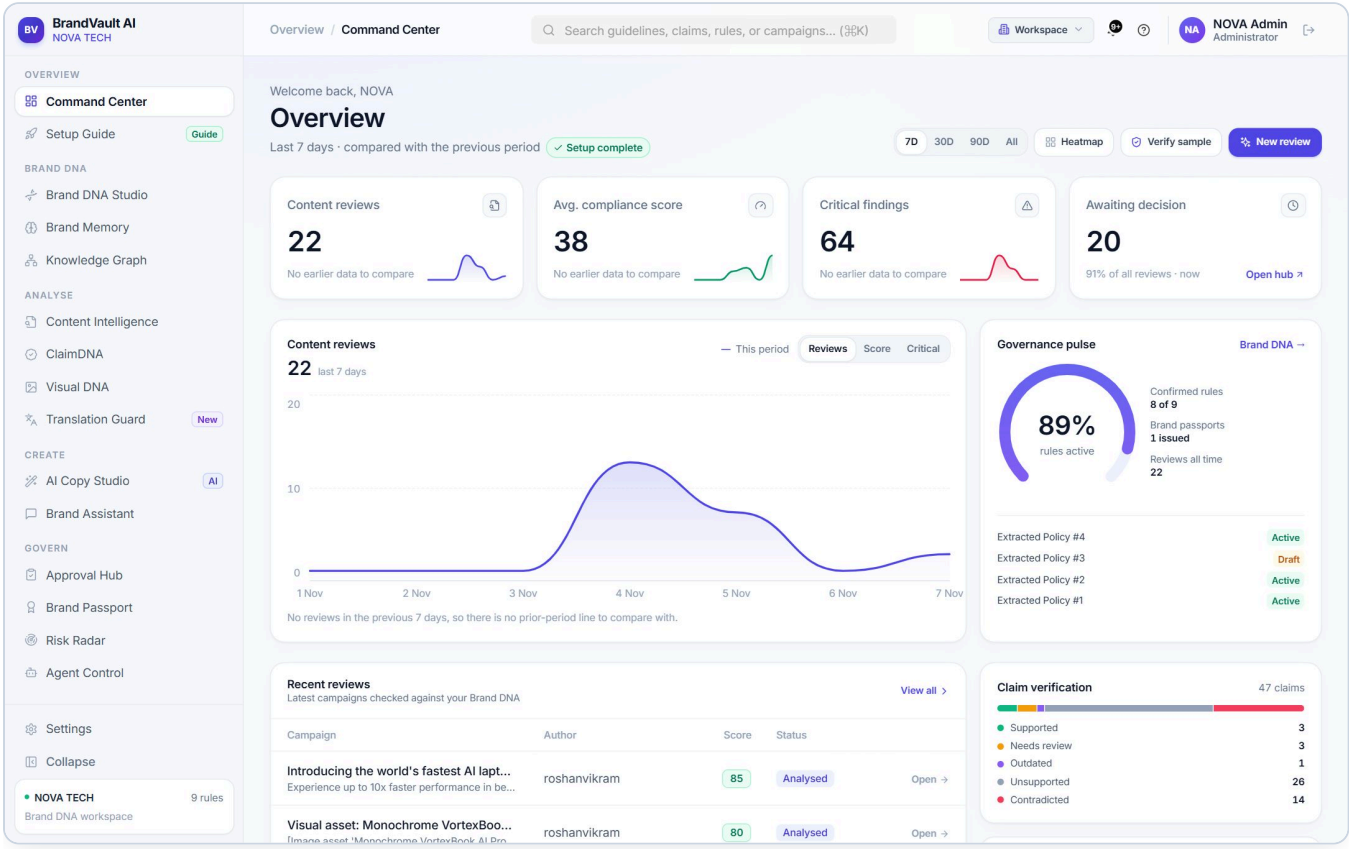


Figure 3.1 — Command Center. Period switcher, KPI cards with trends, review activity chart, governance pulse and recent reviews. The cards say "No earlier data to compare" instead of inventing a trend.

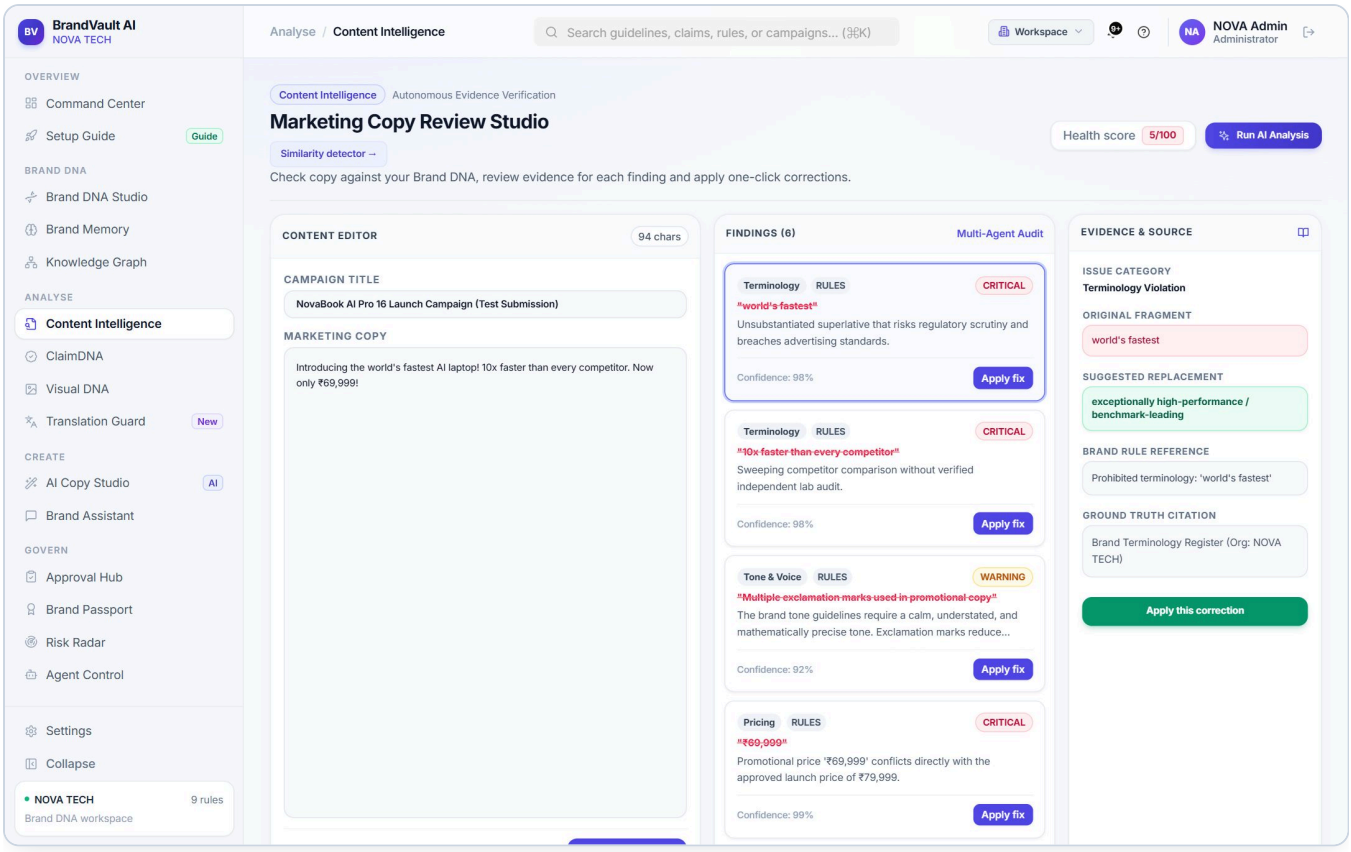


Figure 3.2 — Content Intelligence. The NovaBook AI Pro 16 campaign: each finding shows the flagged text, the rule, the evidence source and a suggested correction that can be applied in one click.

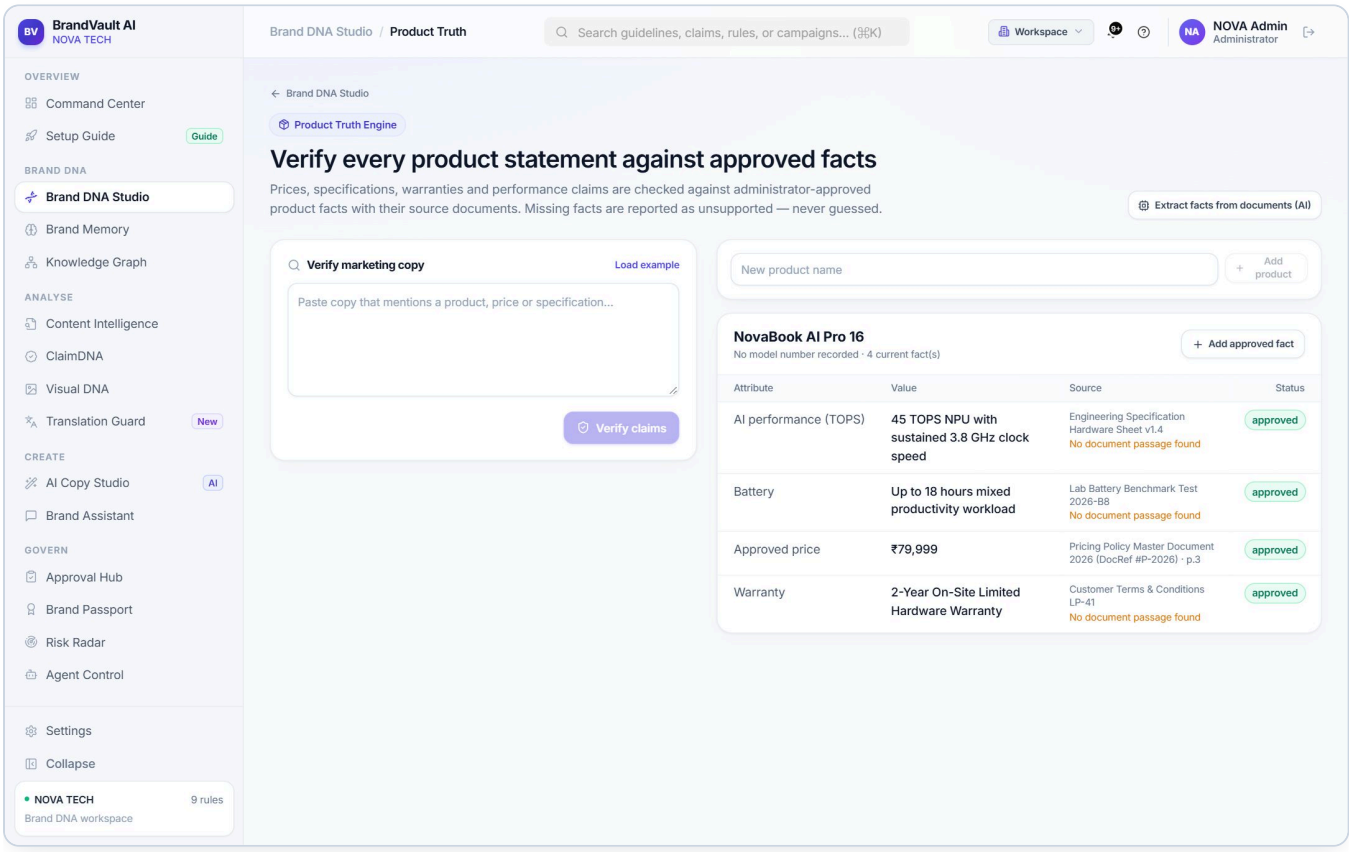


Figure 3.3 — Product Truth Engine. Approved facts for NovaBook AI Pro 16 with source document, page and status. Any copy pasted on the left is verified against these facts.

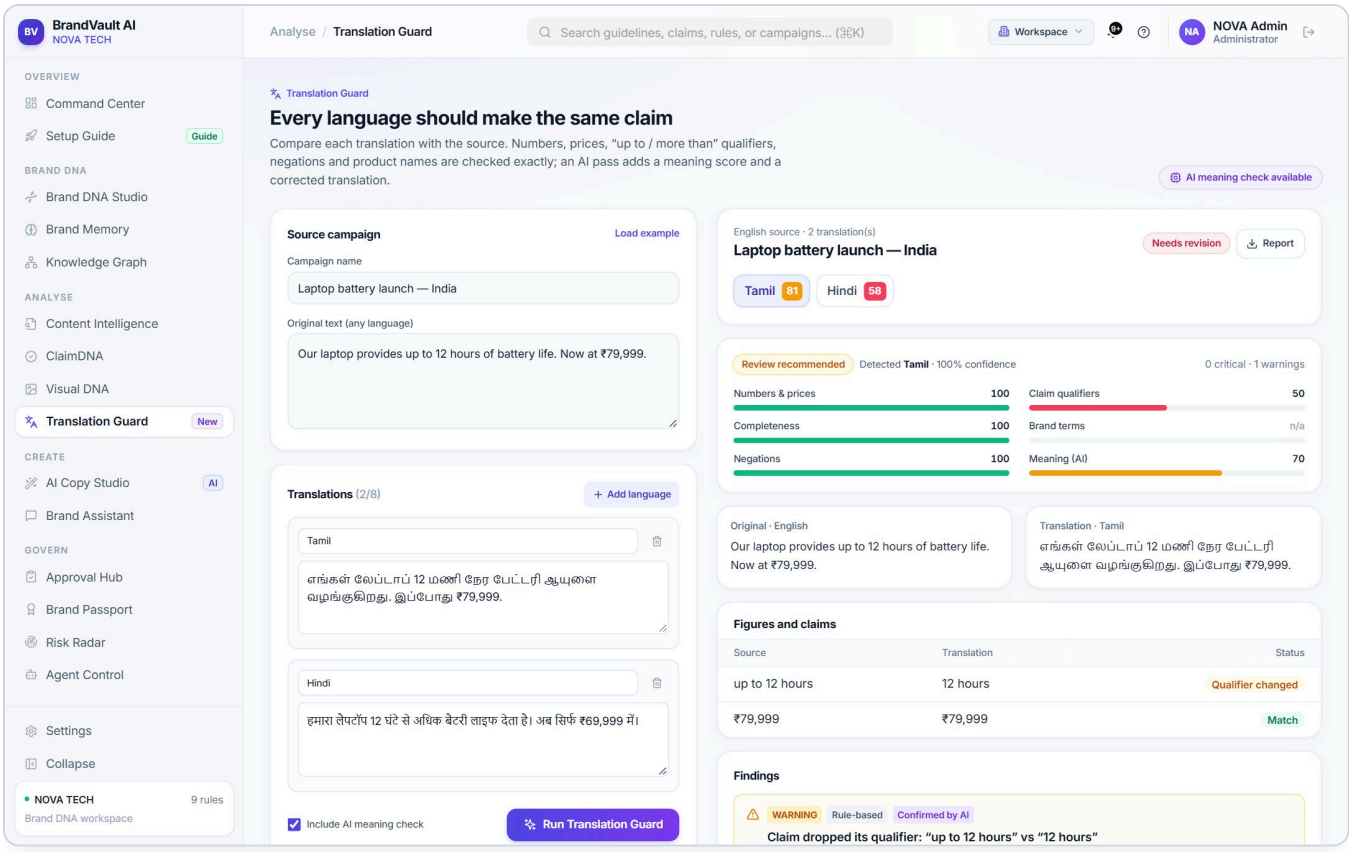


Figure 3.4 — Translation Guard. The Tamil translation dropped the "up to" qualifier ("up to 12 hours" → "12 hours"). The rule-based check flags it, and the AI meaning pass confirms it.

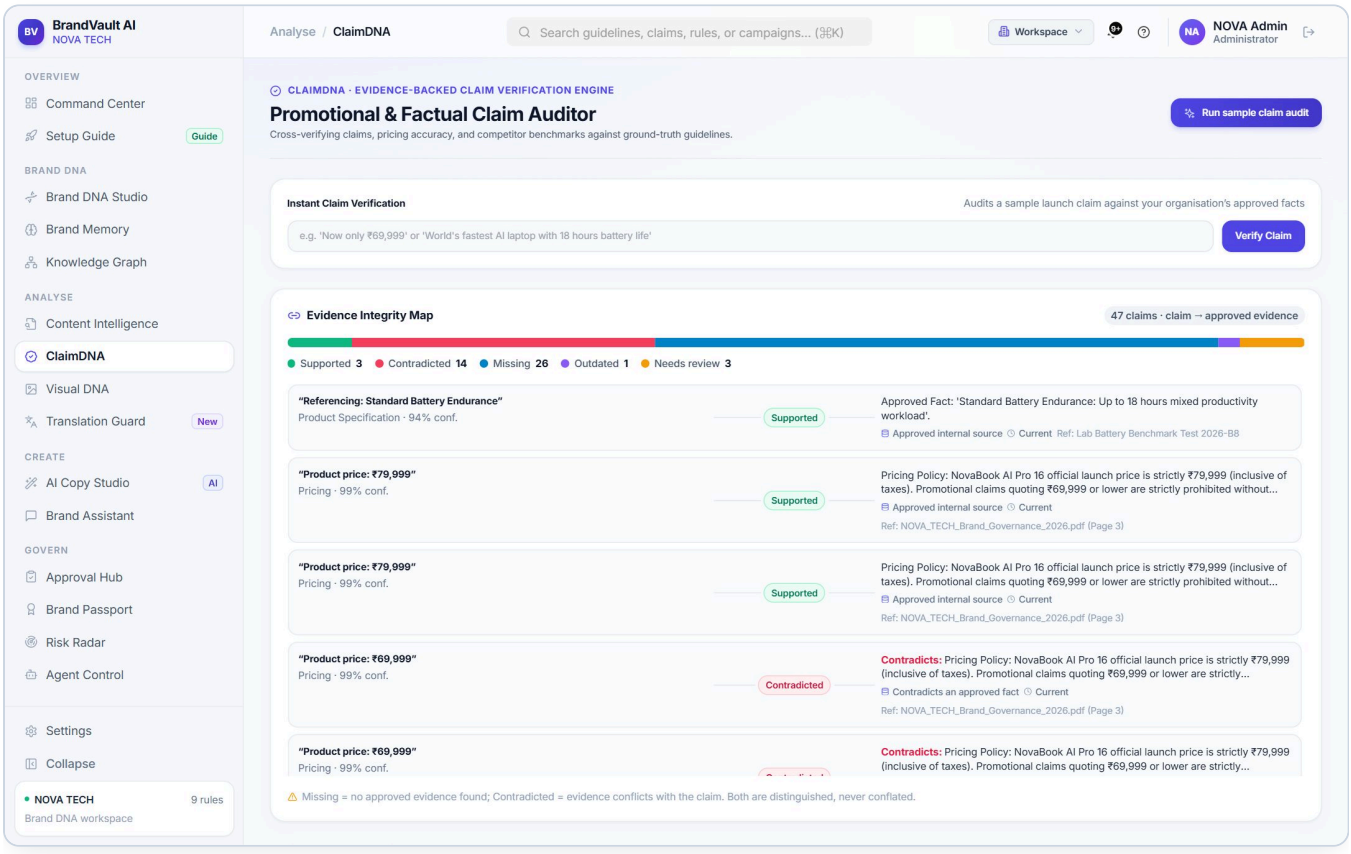


Figure 3.5 — ClaimDNA Evidence Integrity Map. 47 claims classified against approved evidence, with missing evidence kept separate from contradicted claims.

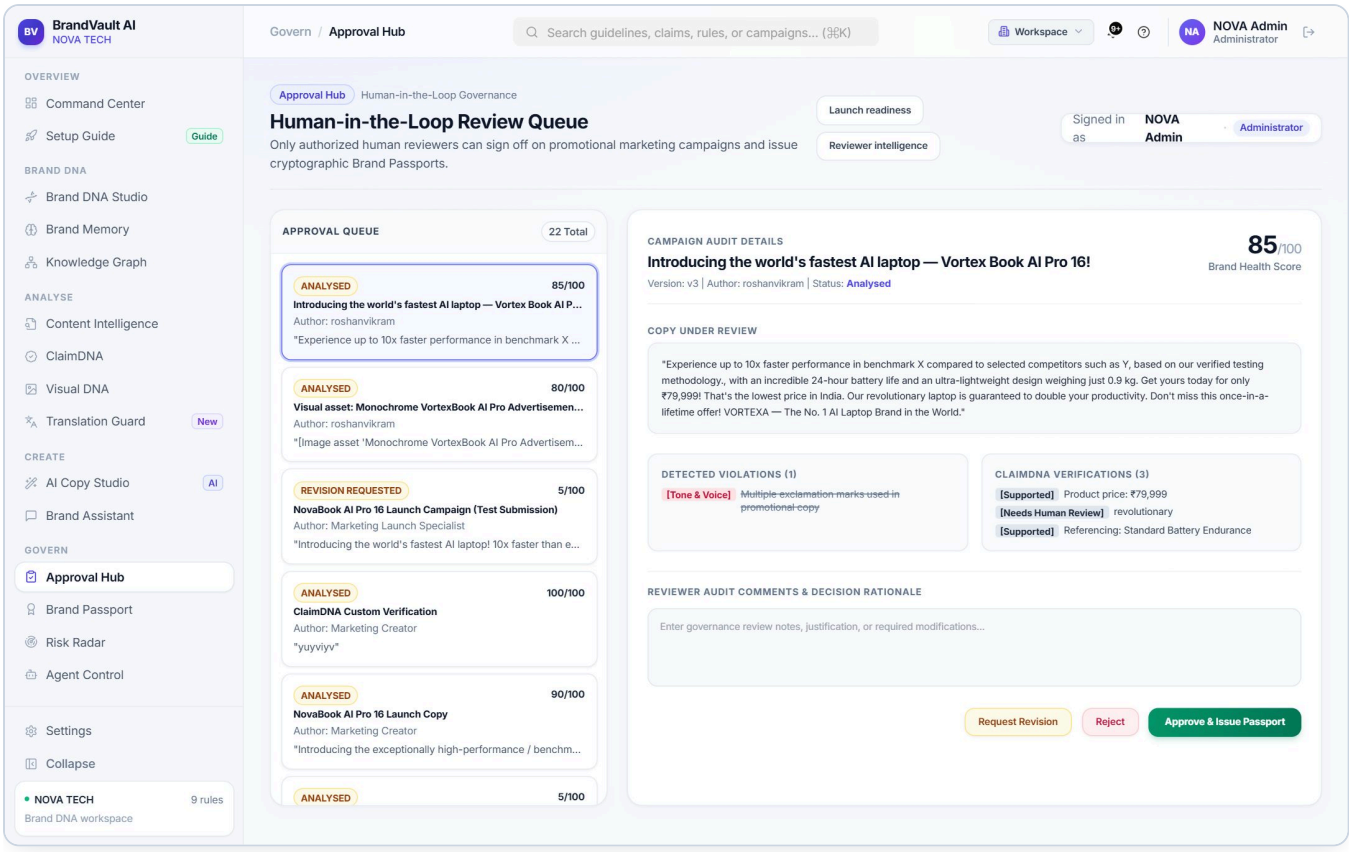


Figure 3.6 — Approval Hub. The human decision gate. A reviewer requests revision, rejects, or approves and issues a Brand Passport for the exact version shown.

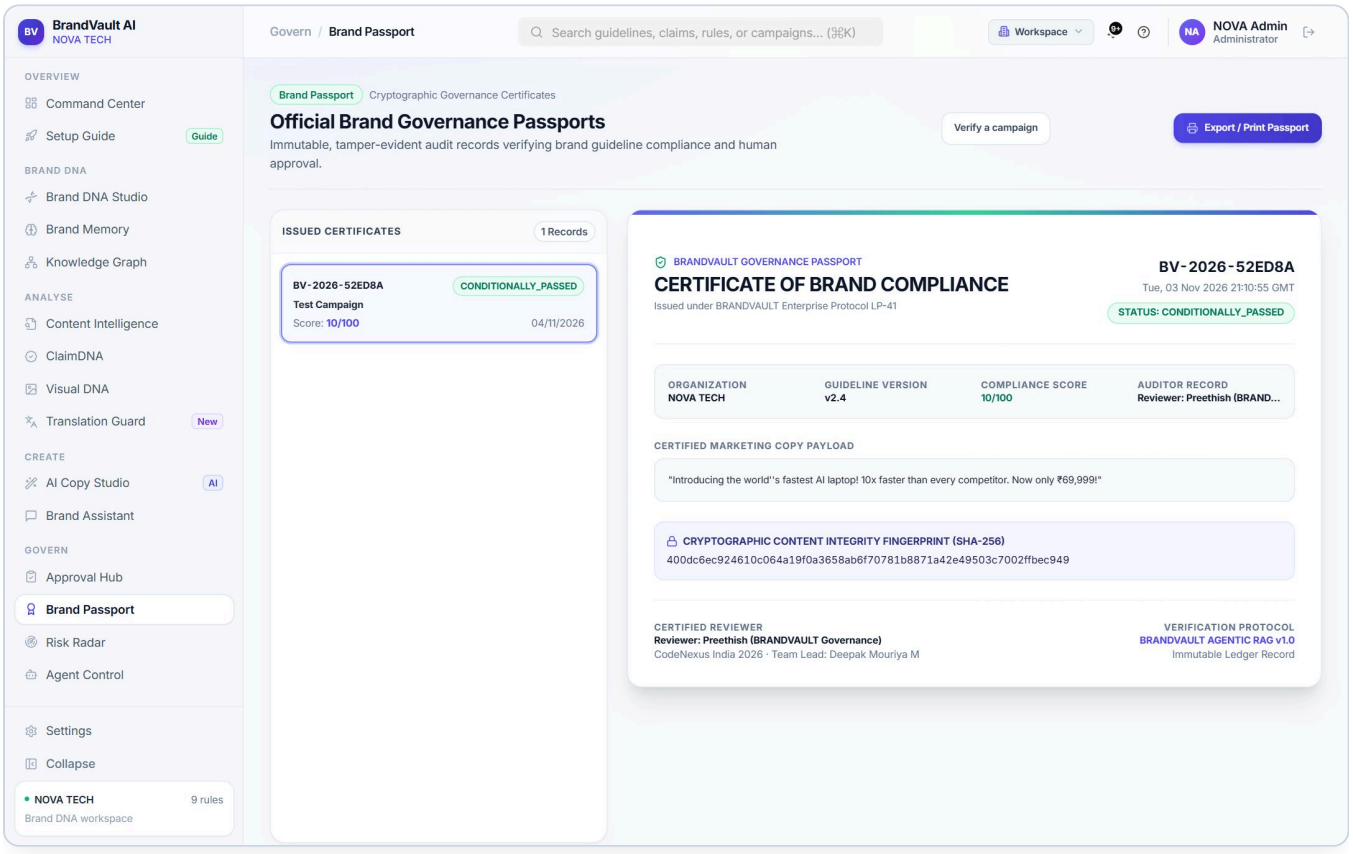


Figure 3.7 — Brand Passport. Certificate of brand compliance with the SHA-256 content fingerprint, guideline version, score and certifying reviewer.

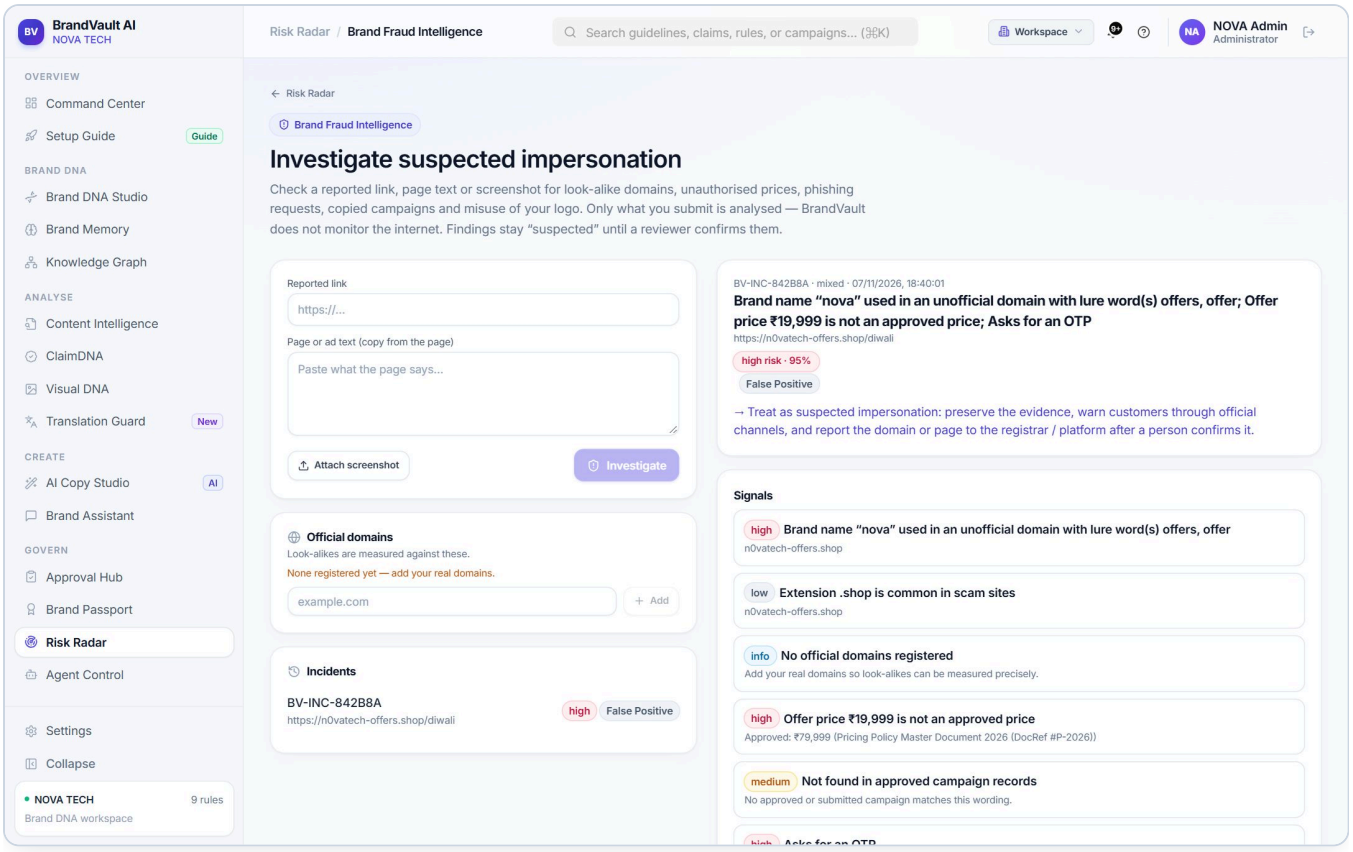


Figure 3.8 — Brand Fraud Intelligence. A test investigation of a look-alike domain. Flags include the brand in an unofficial domain, an unapproved ₹19,999 price and an OTP request. It is recorded as incident BV-INC-842B8A and marked False Positive because it was a test.

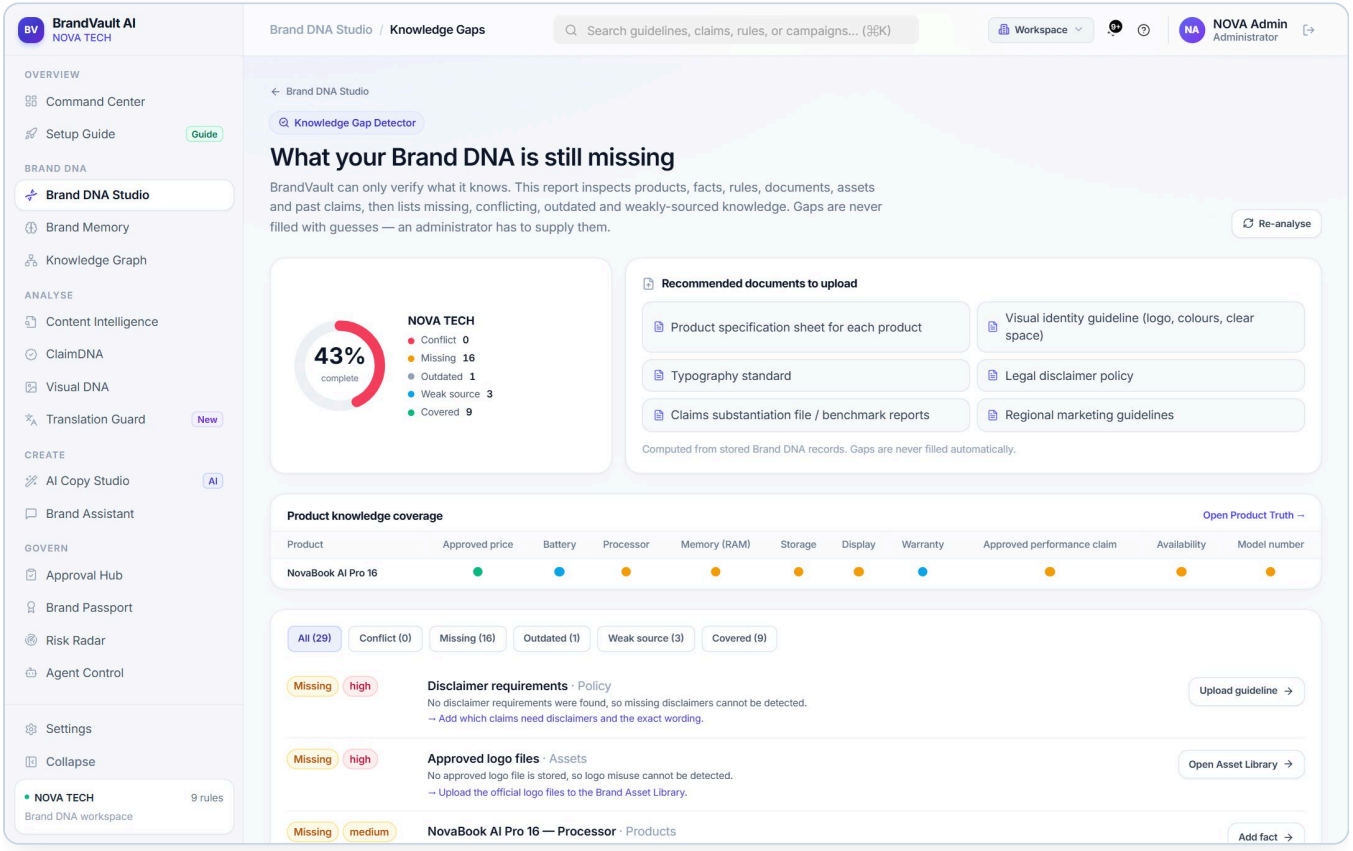


Figure 3.9 — Knowledge Gap Detector. Brand DNA completeness (43%) with prioritised gaps and the documents to upload next.

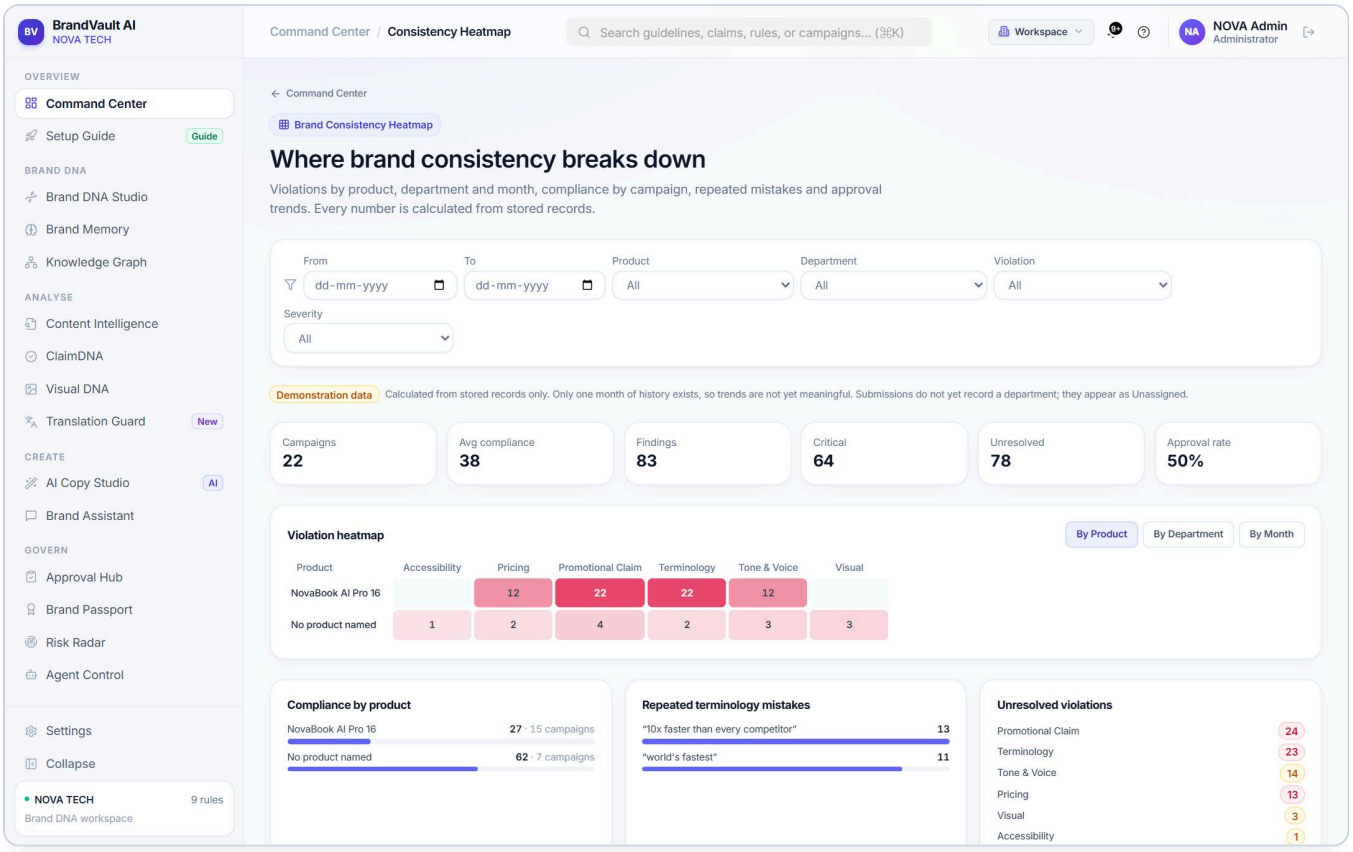


Figure 3.10 — Brand Consistency Heatmap. Violations by product and category, compliance by product, and repeated terminology mistakes. Demo data is labelled on screen.

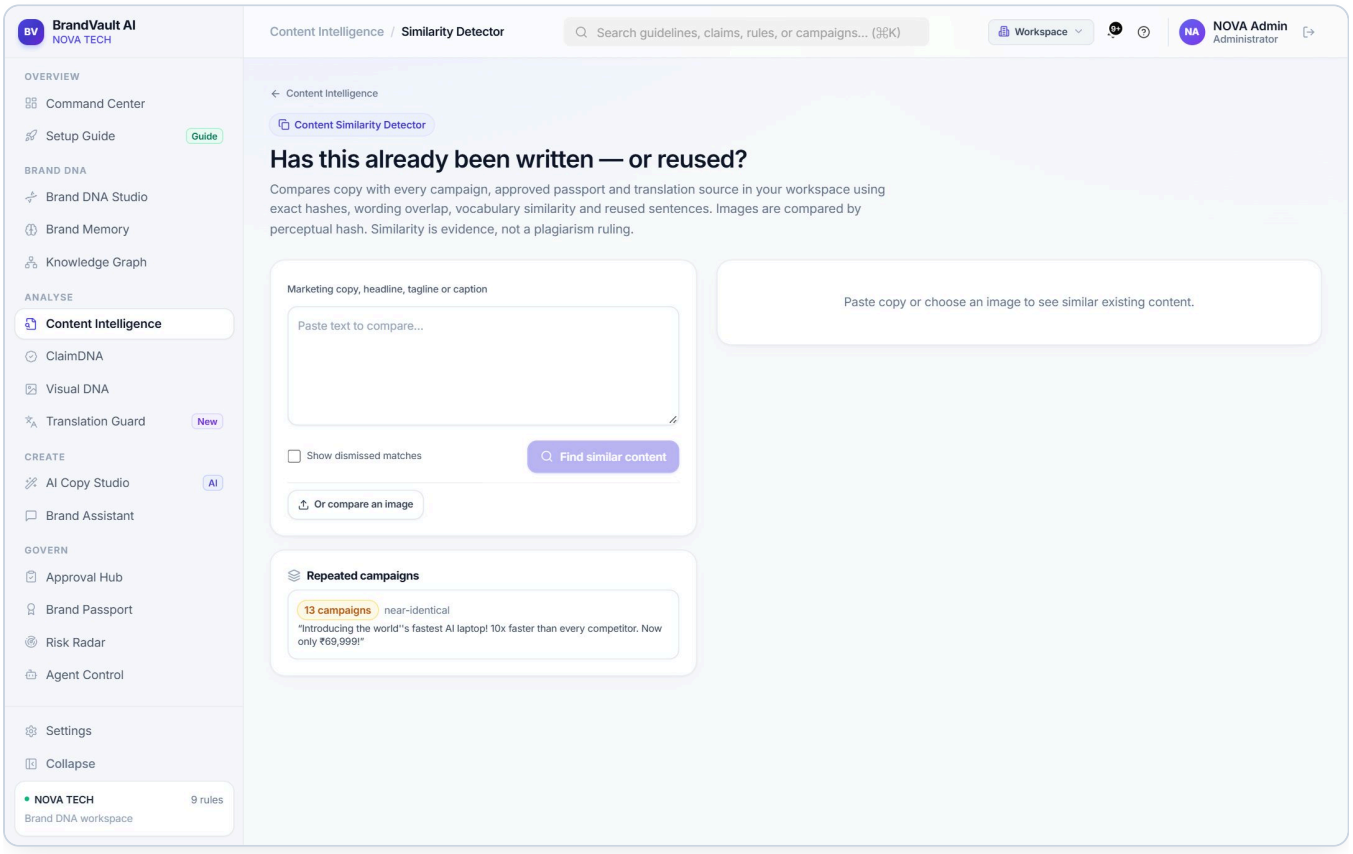


Figure 3.11 — Similarity Detector. Finds repeated campaigns; here a cluster of 13 near-identical NovaBook launch drafts.

04 · REQUIRED ITEM 2

Technology & implementation details

A four-tier architecture: a Next.js web app, a FastAPI backend that enforces identity and tenancy, an intelligence tier that is deterministic first with optional AI, and a relational store ready for PostgreSQL.

PRESENTATION · NEXT.JS 14 (APP ROUTER), REACT 18, TYPESCRIPT, TAILWIND CSS

- 32 screens
- Command Center
- Inspection deck
- Approval gate
- Passport verify
- Custom SVG charts
- Role-aware controls
- Responsive + mobile drawer

▼ HTTPS · Bearer token

BACKEND & ORCHESTRATION · FASTAPI, PYDANTIC V2, SQLALCHEMY 2, PYTHON 3.13

- 100 REST endpoints
- Auth & RBAC
- Tenant resolution
- Rate limiting
- Durable job queue
- Agent run logging
- Audit log
- MCP tool server

▼

INTELLIGENCE TIER · DETERMINISTIC FIRST, AI OPTIONAL AND LABELLED

- Rule engines (terminology, tone, pricing)
- ClaimDNA classifier
- Hybrid retrieval (TF-IDF cosine + keyword)
- Product Truth parser
- Translation Guard
- OpenCV forensics (ORB, homography, pHash, ΔE)
- LLM provider layer: Groq · Anthropic

▼

DATA · SQLITE (DEMO) · POSTGRESQL-PORTABLE MODELS · ALEMBIC SCAFFOLD

- 30 tables
- org_id on every tenant row
- Chunk embeddings
- Version-locked approvals
- Append-only integrity & incident logs
- Evidence file store (hashed)

Implementation highlights

Deterministic first

Prices, numbers, qualifiers, hashes and colour distances are computed exactly. AI only adds findings, and those are labelled.

Swappable AI providers

One provider interface (Groq, Anthropic, none). Timeouts and failures fall back to the deterministic result automatically.

Dry-run analysis

The analyser can run inside a rolled-back transaction, which is how Stress Testing checks hypothetical versions safely.

Technology stack

LAYER	TECHNOLOGIES	WHY WE CHOSE IT
Frontend	Next.js 14.2, React 18, TypeScript 5, Tailwind CSS 3, lucide icons	Fast iteration and typed API contracts. Charts are hand-built SVG, with no chart library.
Backend	FastAPI, Uvicorn, Pydantic v2, SQLAlchemy 2, httpx, pypdf	Typed request/response validation; Pydantic also validates every AI output.
AI models	Groq (`openai/gpt-oss-120b`, used live), Anthropic Claude driver, provider abstraction	Swappable providers. If no key is set, the app runs fully on its deterministic engines.
Computer vision	OpenCV (ORB features, RANSAC homography), NumPy, Pillow	Deterministic, explainable image forensics that runs locally.
Retrieval	L2-normalised TF-IDF vectors persisted per chunk; cosine + keyword hybrid ranking	Runs offline with no paid embedding API, and scores are explainable. Dense embeddings are planned for Round 3.
Data	SQLite for the demo; PostgreSQL-ready ORM types, psycopg2, Alembic	Zero-setup for judges; the same models run on Postgres.
Deployment	Dockerfile, Render blueprint (web + managed Postgres), Procfile, Vercel frontend	Standard, low-cost hosting path (see Known limitations).
Quality	unittest / pytest (172 tests), 36-case evaluation harness	Every engine ships with tests; the evaluation measures precision and recall.

Data model (30 tables)

Identity & brand knowledge

`organizations` · `users` · `brand_guidelines` · `document_chunks` · `brand_rules` · `approved_facts` · `terminology_rules` · `products` · `product_facts` · `brand_assets`

Content & decisions

`submissions` · `revisions` · `findings` · `claim_verifications` · `approval_records` · `brand_passports` · `campaign_integrity_checks` · `launch_readiness_reports` · `launch_check_configs`

Intelligence engines

`multilingual_campaigns` · `campaign_translations` · `asset_investigations` · `simulation_runs` · `similarity_dismissals` · `official_domains` · `fraud_incidents` · `feedback_records`

Operations & audit

`agent_runs` · `jobs` · `audit_logs`

How a campaign is analysed

#	AGENT / STEP	IMPLEMENTATION	TYPE
1	**Terminology Agent**	Matches confirmed terminology rules (e.g. "world's fastest" → "benchmark-leading") and cites the rule.	Deterministic
2	**Brand Voice Agent**	Tone rules such as exclamation density and banned buzzwords.	Deterministic
3	**ClaimDNA Agent**	Extracts price, performance, comparative and freshness claims, retrieves evidence for the organisation, and assigns one of 5 classes. Price checks are exact numeric comparisons.	Deterministic + retrieval
4	**LLM Analyst** (optional)	Finds extra issues. The copy is wrapped as untrusted data, output is Pydantic-validated, merged and de-duplicated, and labelled `llm:<provider>`.	AI, labelled
5	**Suggestion Agent**	Builds the score and one-click corrections from approved replacements.	Deterministic
6	**Engines on demand**	Product Truth, Translation Guard, Launch Readiness, Similarity, Fraud and Forensics each log their own agent run.	Mixed, labelled

Key API groups

GROUP	REPRESENTATIVE ENDPOINTS
Auth	`POST /auth/login · /auth/register · GET /auth/google/callback`
Brand DNA & Product Truth	`POST /brand/guidelines/upload · /products/verify · GET /products/knowledge-gaps`
Analysis & claims	`POST /content/analyze · GET /claim/evidence-map · POST /translation-guard/analyze`
Approval & passport	`POST /approval/decide (expected_version) · POST /passport/verify`
Governance (Phase 2)	`POST /launch/{id}/check · GET /governance/lifecycle · GET /analytics/consistency`
Intelligence (Phase 3)	`POST /simulation/run · /similarity/check · /fraud/investigate · /reputation/upload`

More implementation highlights

Page-accurate citations

PDF text keeps page markers, so every finding cites the exact page of the guideline it came from.

Append-only records

Integrity checks, incident histories and audit events have no update or delete routes.

Additive migrations

New tables are created at startup and new columns are added in place, so existing data survives every upgrade.

The complete endpoint reference is maintained in `API_DOCUMENTATION.md` in the repository.

05 · TESTING & MEASURED QUALITY

Measured, not promised

Round 1 listed target metrics "to be measured". In Round 2 we measured them. The full suite of 172 tests passes, and the analyzer evaluation runs on a labelled dataset in a throwaway database.

Automated test suite

TEST FILE	TESTS
Integration (end-to-end workflow, RBAC)	33
Phase 1 engines (Product Truth, gaps, fingerprint, forensics)	29
Translation Guard	21
Phase 3 intelligence (stress test, similarity, fraud, reputation)	20
Phase 1a security (tenancy, versions, jobs)	19
Phase 2 governance (readiness, lifecycle, heatmap, reviewers)	19
Unit (analyzer, claims, hashing)	18
Visual DNA	8
Configuration	5
Total — all passing	**172**

Analyzer evaluation (36 labelled cases)

1.00 precision

0.86 recall

0.925 F1

ISSUE TYPE	TP	FN	RECALL
Price conflict	6	0	1.00
Terminology	10	0	1.00
Superlative	8	2	0.80
Comparative	4	1	0.80
Tone · Outdated · Needs review	9	0	1.00

Workflow completion: 36/36. No false positives were measured (FP = 0). The 6 misses include 2 visual cases and 1 fact conflict that the text rule engine cannot detect by design.

Engine-level validation

ENGINE	WHAT WE VERIFIED
Asset Forensics	10 labelled synthetic cases (identical, stretched, rotated, cropped, recoloured, outdated, unrelated) were all classified correctly. Verification threshold calibrated at 0.6 (true matches ≥ 0.71, false ≤ 0.48).
Product Truth (AI extraction)	Live run with Groq: 3 facts proposed and duplicates caught. Any fact whose quoted evidence is not verbatim in the source is discarded automatically.
Translation Guard	Live-verified on a Tamil/Hindi example: the dropped "up to" qualifier and a changed price were both caught.
Reputation sentiment	16-item hand-labelled sanity set: 100%. This is a basic check, not a benchmark; a larger labelled set is planned.
Fraud domain analysis	Unit tests for homoglyphs, punycode, typosquats (edit distance ≤ 2), IP hosts and blocked private-network fetches (SSRF).

06 · SECURITY & RESPONSIBLE AI

Built so AI never has the final word

Tenant isolation

Every tenant-owned query is filtered by the organisation resolved from the signed-in identity. A record from another organisation is rejected, and this is tested for every engine.

Role-based access

Administrators manage Brand DNA, products, domains and policies. Reviewers decide. Creators submit. AI agents hold no approval rights.

Version-locked approvals

An approval records the version and content hash. Any edit creates a new version that must be re-approved, and the old passport is superseded.

Prompt-injection defence

Submitted copy is wrapped as untrusted data. Model outputs must pass Pydantic schemas, and agents have no tools that can approve or publish.

Safe external fetching

Fraud URL fetching is off by default. When enabled it allows public hosts only, follows no redirects and caps size and time (SSRF guard).

Secrets & abuse limits

The secret key is required in production. CORS is driven by environment settings, and sign-in and AI routes are rate-limited. Uploads are type- and size-checked.

Responsible AI commitments, as implemented

- **Absence of evidence is not falsity.** Missing evidence is shown as "Unsupported", never "False".
- **AI output is always labelled** with provider and model, and the deterministic result stays authoritative.
- **Suggestions never auto-apply.** Brand Memory and Reviewer Intelligence only create drafts that an administrator must confirm.
- **Hypotheticals stay hypothetical.** Stress tests run in dry-run mode and are labelled; the real campaign is untouched.
- **Honest empty states.** Thin data (2 human decisions, one month of history) is stated on screen instead of being padded with invented numbers.

07 · REQUIRED ITEM 3

Changes made to the original plan

The Round 1 proposal described a 24-hour MVP. We kept every governance principle and the full workflow, and changed some implementation choices so the product works reliably on any judge's machine. The table compares what was planned with what was built and gives the reason for each change.

AREA	ROUND 1 PLAN	WHAT WE BUILT	WHY WE CHANGED IT
Database	PostgreSQL 16 + pgvector from day one	SQLite for the demo; models are Postgres-portable, with Alembic and a Render blueprint	Zero setup for judges and offline demos. Moving to PostgreSQL is a config change, scheduled for Round 3.
Embeddings	1536-d dense vectors, HNSW index, BM25 + RRF, reranker	Persisted TF-IDF vectors with cosine + keyword hybrid ranking and status/tenant filters	Works without a paid embedding API, is deterministic and explainable, and was fast enough for the demo corpus.
LLM provider	Claude API only	Provider layer with Anthropic and Groq drivers (`openai/gpt-oss-120b` used live) and a full deterministic fallback	Availability and cost during the build. The product never depends on one vendor.
OCR	Tesseract text extraction from images	Not shipped; the app reports "OCR unavailable" honestly	The OCR binary was not available in our build environment, and we chose not to fake results. Scheduled for Round 3.
Visual DNA	OpenCV clear space, k-means palette, Claude Vision	Measured palette and contrast, plus a new OpenCV Asset Forensics engine (logo localisation, ΔE, pHash)	We prioritised logo misuse detection, the higher-risk problem. Clear-space checks move to Round 3.
MCP	MCP server with 8 governance tools over JSON-RPC	7 governance tool functions in `mcp_server.py` with a self-test; protocol transport not yet wired	Engine depth took priority. A full MCP transport is part of Round 3 improvement 2.
Agents	Orchestrator + 7 specialist agents	Sequential coordinator with Terminology, Voice, ClaimDNA, LLM Analyst and Suggestion agents, plus 8 engine agents; all runs logged	A sequential pipeline is easier to audit. Channel Adapter and Regulatory agents (R1 stretch) move to Round 3.
Voice Copilot (stretch)	Whisper speech-to-text + text-to-speech	A grounded, text-based Brand Assistant with citations; the product tour reads aloud	Grounded text answers were more useful to reviewers than voice. Dropped per the Round 1 cut line.
Job queue	Celery	Database-backed durable jobs	Fewer moving parts and jobs survive restarts, with no Redis or broker to run.
Row-level security (stretch)	PostgreSQL RLS policies	Application-level tenant isolation on every query, covered by tests	RLS needs PostgreSQL. It will be added on top of app-level isolation in Round 3.
Scope	10 views, MVP workflow	32 screens, 12 advanced engines, Translation Guard and 5 extra modules	The core finished early, so we invested in engines that make governance proactive.
Build process	Single 24-hour sprint	Phased: Phase 0 audit → 1a security → engine phases 1–3, each with tests and docs	An audit found security gaps (tenancy, approval versioning) that had to be fixed before adding features.

08 · REQUIRED ITEM 4

Three major improvements for Round 3

Each improvement adds a new capability to the product, not a visual change. Each closes a real gap from Round 1, has a clear technical plan, and comes with measurable success criteria.

1 Multimodal Creative Inspection

Read and verify the text inside images, banners, PDF decks and video frames, not just typed copy.

Most marketing content is visual, but today only typed copy is fully checked. A banner that says "₹69,999" passes untouched because the price is inside the image.

WHAT WE WILL BUILD

- **OCR pipeline** (Tesseract with a vision-model fallback) extracts text with bounding boxes from PNG, JPEG, WebP and PDF pages.
- Extracted text flows into **ClaimDNA, Product Truth and Terminology**, so findings on images get the same evidence cards.
- **Logo clear-space and placement rules** with OpenCV, reusing Asset Forensics localisation.
- **Region overlays** on the creative showing exactly where each issue is.
- **Video and deck support:** sample key frames and slides, then inspect each one.

HOW WE WILL MEASURE IT

- A labelled set of 25+ creatives (prices, superlatives, logo misuse).
- Text-extraction accuracy and claim precision/recall reported in the product, not just targeted.
- Closes the 2 visual misses in today's evaluation.

BUILDS ON

- Asset Forensics, ClaimDNA, Product Truth, Visual DNA.

2 Publishing Gate & Channel Integrations

Check content where it is written and block publishing without a valid Brand Passport.

Governance only works if people use it. Today a team must open BrandVault and paste copy in. Round 3 brings the check into the tools where content is created and published.

WHAT WE WILL BUILD

- **Public Check API** with per-organisation API keys and signed webhooks (analysis complete, approved, passport revoked).
- **Publishing gate:** a CMS/CI step that refuses to publish content whose fingerprint does not match a valid passport, reusing Campaign DNA Fingerprint.
- **Channel Adapter agent** (from the Round 1 stretch plan): character limits, aspect ratios and required elements per channel.
- **Regulatory disclaimer agent:** region- and category-specific mandatory disclaimers.
- **Full MCP transport** (JSON-RPC) so external AI agents and IDEs can call our read-only governance tools.
- **Slack / email notifications** for review requests and expiring approvals.

HOW WE WILL MEASURE IT

- End-to-end demo: a draft written in an external tool is checked, approved and published only after passport verification.
- Permission enforcement: 100% of unauthorised API and MCP calls blocked (Round 1 metric, now testable).

BUILDS ON

- Brand Passport, Launch Readiness, MCP tool server, roles.

3 Semantic Evidence Engine on PostgreSQL + pgvector

Better retrieval, more document types and a live, production-grade deployment.

Keyword-weighted retrieval misses paraphrases ("all-day battery" vs "18 hours"), and only PDF guidelines can be ingested. This improvement delivers the retrieval design from Round 1 and puts the product online.

WHAT WE WILL BUILD

- **Dense embeddings in pgvector** with HNSW, fused with BM25 via Reciprocal Rank Fusion, and an optional cross-encoder reranker.
- **New ingestion formats:** DOCX, HTML and spreadsheets (price lists), with page and section citations.
- **Multi-hop evidence** across document versions, so "Outdated" is decided from version history.
- **Guideline Gap Agent** made real: finds rules that contradict each other across documents.
- **Live deployment** on managed PostgreSQL with row-level security on top of app isolation, and demo accounts removed.

HOW WE WILL MEASURE IT

- Retrieval recall@3 and evidence-attribution accuracy on a labelled claim set, compared with today's engine.
- Recall above today's 0.86 with precision kept at or near 1.00.
- A public URL judges can use without installing anything.

BUILDS ON

- RAG retriever, ClaimDNA, Knowledge Gap Detector, deployment config.

Round 3 delivery plan

IMPROVEMENT	PLANNED OWNERS	MILESTONES
1 · Multimodal Creative Inspection	Aishwarya R M (lead), Roshan Vikram AV	OCR service with boxes → claims and terminology on extracted text → clear-space rules → region overlays → labelled creative set and measured accuracy.
2 · Publishing Gate & Integrations	Deepak Mouriya M (lead), Pritheesh Kumar V	API keys and webhooks → passport-verified publishing step → Channel Adapter and disclaimer agents → MCP JSON-RPC transport → notifications.
3 · Semantic Evidence Engine	Pritheesh Kumar V (lead), Roshan Vikram AV	PostgreSQL + pgvector migration → dense embeddings with BM25/RRF → DOCX/HTML ingestion → version-aware evidence → live deployment with RLS.

Same rules as Round 2: every improvement ships with tests, measured results and honest on-screen limits. No AI agent gains the right to approve or publish.

09 · DEMO SCRIPT & HOW TO RUN

A five-minute judge walkthrough

TIME	STEP	WHAT TO SHOW
0:00	**Command Center**	Real KPIs, trend chart and governance pulse for the NOVA TECH workspace.
0:40	**Content Intelligence**	Open the NovaBook launch campaign: ₹69,999 contradicts the approved ₹79,999, "world's fastest" is unsupported. Apply a correction.
1:30	**Product Truth**	Paste copy with a wrong battery claim and see it contradicted with source page and excerpt.
2:10	**Translation Guard**	The Tamil version drops "up to"; the rule check and the AI both flag it.
2:50	**Launch Readiness → Approval**	Run the 12 checks, then approve as a Reviewer. A Brand Passport is issued with SHA-256.
3:40	**Passport verify**	Paste an edited copy; it shows "Modified After Approval" with a word diff.
4:20	**Fraud Intelligence**	Investigate a look-alike link with a fake price and an OTP request, which opens a numbered incident.

Run it locally

Backend

```
python -m venv backend/venv
backend/venv/Scripts/pip install -r
backend/requirements.txt
backend/venv/Scripts/python -m uvicorn
backend.main:app --port 8000
```

Frontend

```
cd frontend
npm install
npm run dev  # http://localhost:3000
```

Demo accounts for the NOVA TECH workspace (Administrator, Reviewer, Creator) are seeded automatically and listed in `DEPLOYMENT.md`. They are for judging only. An AI key (`GROQ_API_KEY` or `ANTHROPIC_API_KEY`) is optional; without one, every engine runs on its deterministic path.

10 · KNOWN LIMITATIONS

What is not done yet

We list limitations openly so judges can trust everything else in this report.

LIMITATION	DETAIL	PLANNED FIX
Not publicly deployed	Deployment files exist but have not been executed on a host; the app runs locally.	Round 3 · #3
No OCR	Text inside images is not read; the UI says so.	Round 3 · #1
PDF-only ingestion	DOCX and HTML guidelines are not yet supported.	Round 3 · #3
Retrieval	Sparse (TF-IDF) vectors, so paraphrased evidence can be missed.	Round 3 · #3
MCP transport	Tool functions exist; the JSON-RPC protocol layer is not wired.	Round 3 · #2
Fraud monitoring	Only submitted links, text and screenshots are analysed; there is no internet monitoring.	By design
Sentiment	Lexicon-based estimate; validated only on a small sanity set.	Larger labelled set
Thin history	The demo workspace has 2 human decisions and one month of data, so trend and reviewer insights are limited. This is stated in the UI.	Grows with use
Notifications	Alerts show on screen only; no email or push yet.	Round 3 · #2
Rate limiting	In-process, so limits are not shared across multiple server instances.	With deployment

11 · CONCLUSION

From verification to proactive governance

In Round 1 we argued that the bottleneck in enterprise marketing has shifted from producing content to verifying it. In Round 2 we built the platform to address it. Every campaign is checked against confirmed brand knowledge, and every finding carries its evidence. Every approval is tied to the exact version a person reviewed, and every passport can be verified later.

The 12 advanced engines move BrandVault from reactive checking to proactive governance. They find gaps in brand knowledge before campaigns are written, stress-test changes before they are made, catch translation drift, spot impersonation and track the lifecycle of every approved asset.

Round 3 completes the picture: inspecting what is inside images and videos, checking content inside the tools teams already use, and putting a semantic evidence engine on production infrastructure.

Agents advise
Specialist agents find issues and cite evidence, and never act on their own.

Evidence grounds
Every finding links to a confirmed rule, fact or source page.

Humans decide
Only authorised people approve, and every decision is recorded.